

Trees overview

What is tree ?

- A data structure with nodes and their connection
- Basic tree node contains:
 - A value
 - An array of children connections (0 or many)

Where are tree ?

- Your filesystem is a tree
- The dom is a tree
- Trees are massively important in compilers. You have probably minimally heard the term Abstract Syntax Tree.
- and there is more...

Some terminology

- root - the most parent node. The First. Adam.
- height - The longest path from the root to the most child node
- binary tree - a tree in which has at most 2 children, at least 0 children
- general tree - a tree with 0 or more children
- binary search tree - a tree in which has a specific ordering to the nodes and at most 2 children
- leaves - a node without children
- balanced - A tree is *perfectly* balanced when any node's left and right children have the same height.
- branching factor - the amount of children a tree has.